

03 – SQL: Structured Query Language

Define and manipulate data in your database

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OUTLINE

0) Recall

1) What is SQL?

2) Installing software

3) Data Definition Language (DDL)

4) Data Manipulation Language (DML)

På svenska:



In English:

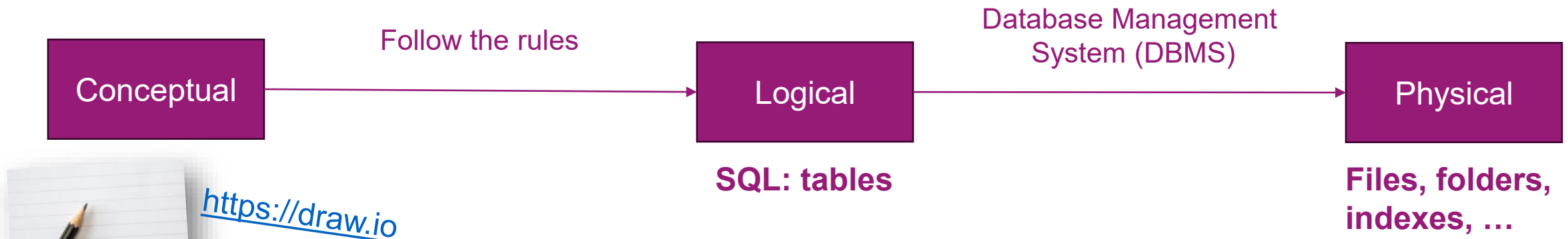


0) RECALL



0) RECALL

- To create a Relational Database, the traditional way is to use the **Entity-Relationship (ER) Model** which has three levels:
 - The **conceptual** model: a diagram representing the entities and their relationships
 - The **logical** model: a list of tables
 - The **physical** model: the organization of the information on the computer hard drive (files)



Today, we enter the **LOGICAL MODEL: TABLES**

1) WHAT IS SQL?

SQL (Structured Query Language) is a **standardized programming language** used for managing and manipulating relational databases [1][2][3]. It allows users to store, retrieve, update, and delete data in database management systems [5].

SQL was developed in the 1970s and has since become the **global standard** for relational database management systems (RDBMS) [4]. It is widely used by database administrators, developers, and data analysts for various tasks, including:

1. Executing queries to **retrieve** data
2. **Inserting** new records into databases
3. **Updating** existing records
4. **Deleting** records from databases
5. **Creating** new databases and tables
6. Setting **permissions** on database objects

SQL is particularly useful for handling structured data and managing relationships between different data entities [3]. It operates on a **declarative model**, meaning **users describe what they want to achieve rather than specifying how to do it** [4].

1) WHAT IS SQL?

Key features of SQL include:

- **Standardization**: SQL became an ANSI standard in 1986 and an ISO standard in 1987 [2][3].
- **Versatility**: It can be used across various database systems, including MySQL, Oracle, and Microsoft SQL Server [5].
- **Scalability**: SQL can handle growing data needs effectively [4].
- **Data manipulation**: It excels at processing and analyzing large datasets [4].

While SQL has a standardized core, different database systems may have their own dialects or extensions, such as T-SQL for MS SQL Server and PL/SQL for Oracle [4].

Citations:

- [1] <https://www.geeksforgeeks.org/what-is-sql/> [2] https://www.w3schools.com/sql/sql_intro.asp [3] <https://en.wikipedia.org/?title=SQL>
[4] <https://www.ibm.com/think/topics/structured-query-language> [5] <https://www.techtarget.com/searchdatamanagement/definition/SQL>
[6] <https://www.atlassian.com/data/sql> [7] https://www.reddit.com/r/AskProgramming/comments/16o4ijx/this_may_sound_a_bit_stupid_but_what_exactly_is/
[8] <https://www.youtube.com/watch?v=27axs9dO7AE> Answer from Perplexity: pplx.ai/share

1) WHAT IS SQL?

Structured Query Language (SQL) is the relational databases

Data Definition Language (DDL)

and

Data Manipulation Language (DML).

It is a **language**, you must **learn** its **syntax** to be able to use it!

1) WHAT IS SQL?

Data Definition Language (DDL) is a subset of **SQL** used to define and manage the structure of database objects [1][2]. It provides commands for creating, modifying, and deleting database elements such as tables, indexes, views, and sequences [2][3].

- Key Aspects of DDL:

1. **Purpose:** DDL is used to describe, comment on, and label database objects, as well as impose or drop constraints on tables [2].
2. **Common Commands:** The most frequently used DDL commands are CREATE, ALTER, and DROP [1]. For example:
 - **CREATE:** Used to create new database objects like tables or indexes [1].
 - **ALTER:** Modifies existing database structures [1].
 - **DROP:** Removes database objects [1].
3. **Scope:** DDL deals with the structure of the database, not the data itself [1]. It affects the schema of the database rather than the content [3].
4. **Immediate Effect:** When a DDL statement is executed, it takes effect immediately in the database [1].
5. **User Access:** Typically, DDL commands are not used by general application users but by database administrators or developers with appropriate permissions [1].

DDL plays a crucial role in database management by providing the tools necessary to define and maintain the structural integrity of databases, ensuring efficient data storage and retrieval[5].

Citations:

[1] <https://www.techtarget.com/whatis/definition/Data-Definition-Language-DDL>

[3] https://en.wikipedia.org/wiki/Data_definition_language

[5] <https://www.lenovo.com/us/en/glossary/what-is-ddl/>

[7] <https://celerddata.com/glossary/data-definition-language-ddl>

[2] <https://www.secodia.co/glossary/what-is-a-data-definition-language-ddl>

[4] <https://www.getdbt.com/blog/guide-to-ddl>

[6] <https://www.techopedia.com/definition/1175/data-definition-language-ddl>

Answer from Perplexity: pplx.ai/share

1) WHAT IS SQL?

Data Manipulation Language (DML) is a subset of SQL used for managing and manipulating data within a database. It focuses on interacting with the actual data stored in database tables, allowing users to perform CRUD (Create, Read, Update, Delete) operations [1][3].

- Key DML Commands:

1. **SELECT**: Retrieves data from one or more tables [5]. Data Retrieval: Fetching specific data based on certain criteria.
2. **INSERT**: Adds new records to a table [3][5]. Data Entry: Inserting new records into database tables.
3. **UPDATE**: Modifies existing data in a table [3][5]. Data Modification: Updating existing records with new information.
4. **DELETE**: Removes records from a table [3][5]. Data Removal: Deleting unnecessary or outdated records.

- Characteristics of DML:

- DML statements are executed immediately and can be undone with a rollback statement [2].
- They are typically used by end-users, applications, or systems interacting with the database [2].
- DML operations affect the data within tables, not the database structure itself [4].

DML plays a crucial role in database management by providing the tools necessary for efficient data handling and retrieval, ensuring that databases remain up-to-date and relevant for various applications and business needs [3][7].

Citations:

- [1] <https://www.datasunrise.com/knowledge-center/dml-data-manipulation-language/>
- [3] <https://satoricyber.com/glossary/dml-data-manipulation-language/>
- [5] https://en.wikipedia.org/wiki/Data_manipulation_language
- [7] <https://staragile.com/blog/data-manipulation-language>

- [2] <https://appmaster.io/blog/difference-between-ddl-and-dml>
- [4] <https://www.javatpoint.com/ddl-vs-dml>
- [6] <https://stackoverflow.com/questions/2578194/what-are-ddl-and-dml>
- [8] <https://www.scaler.com/topics/ddl-dml-dcl/>

Answer from Perplexity: pplx.ai/share

1) WHAT IS SQL?



1) WHAT IS SQL?



1) WHAT IS SQL?



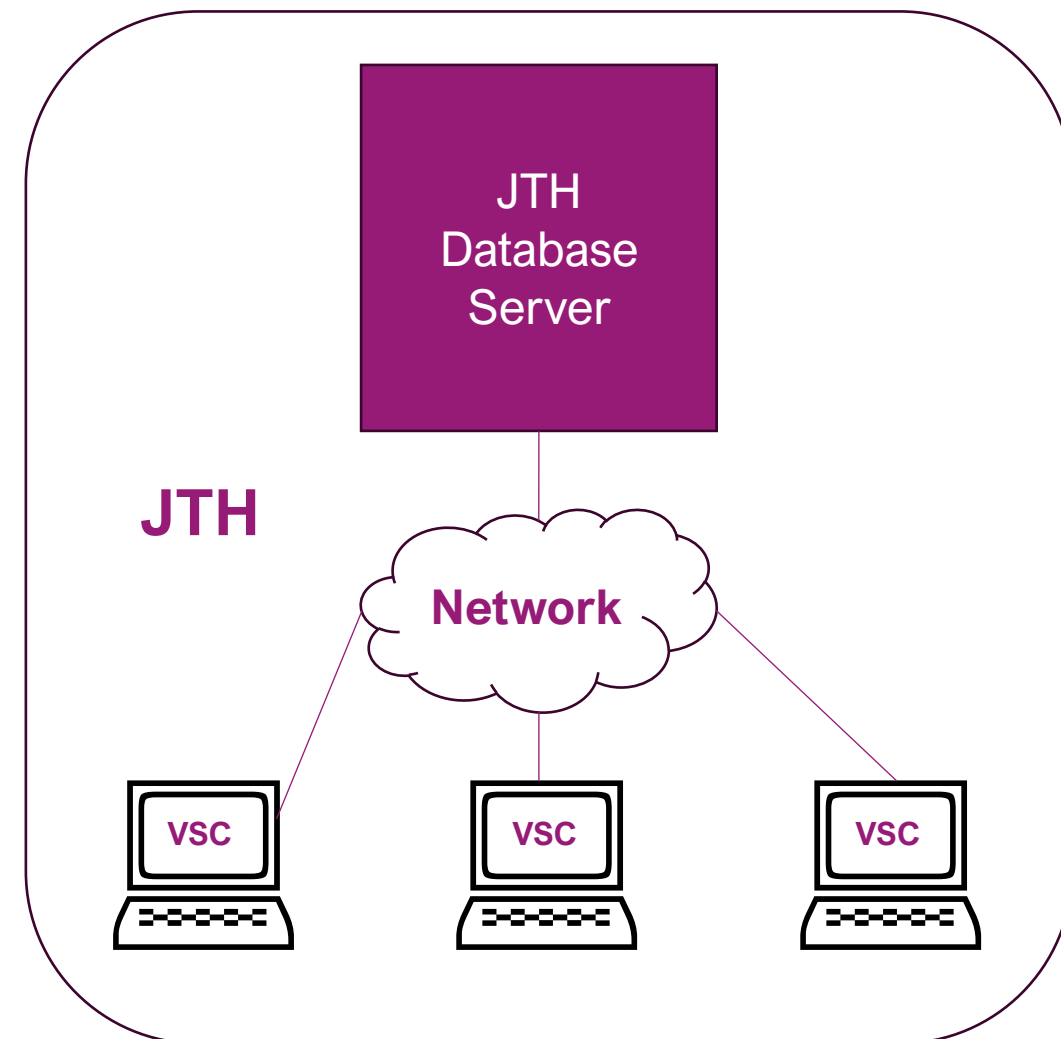
Structured **Q**uery **L**anguage

2) INSTALLING SOFTWARE

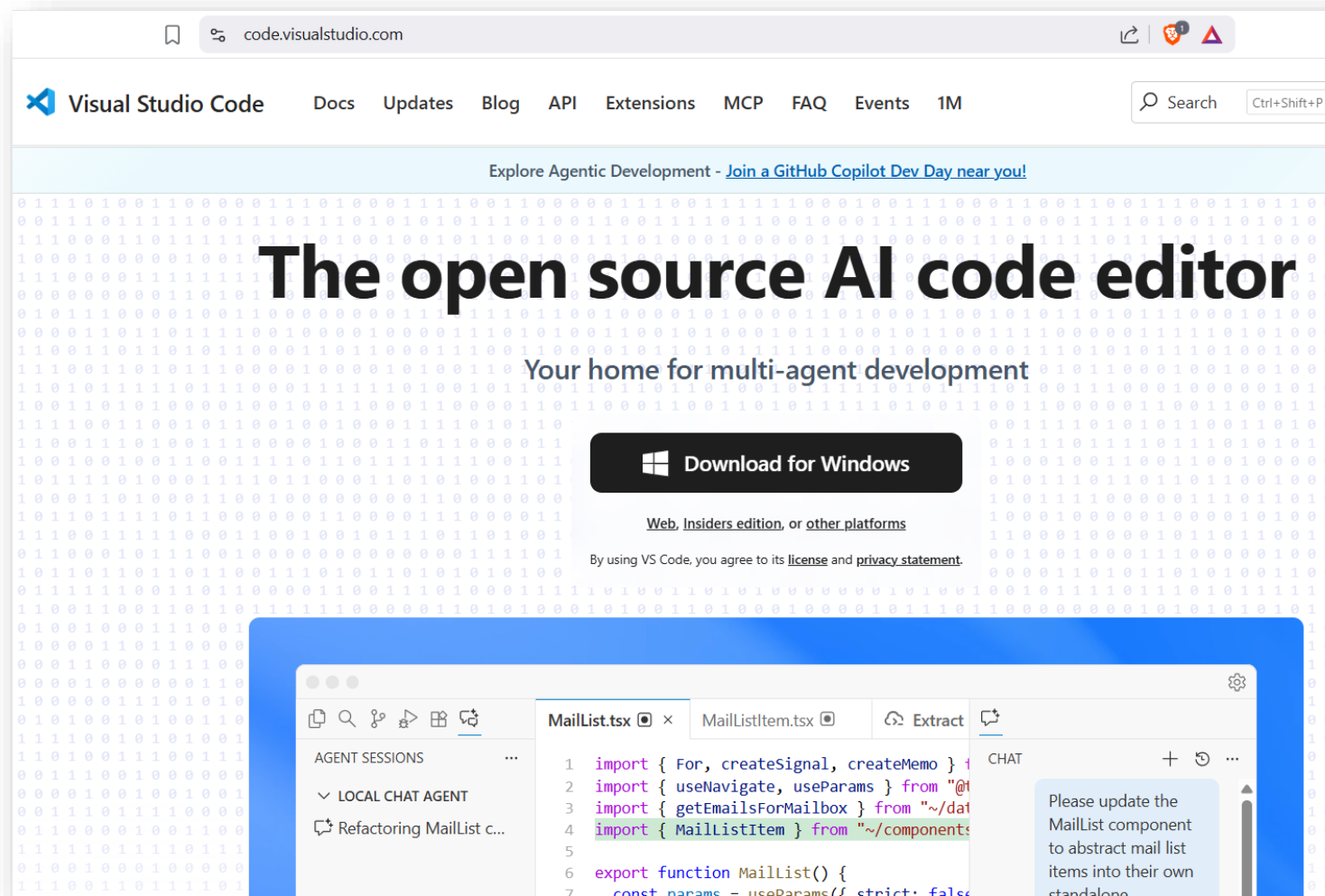
In this course, we use Microsoft SQL Server (MSSQL) as our main **Relational Database Management System (DBMS)**

To connect to MSSQL, we will need to download and install **Microsoft Studio code + Microsoft MSSQL extension**

JTH: Jönköping Tekniska Högskolan,
School of Engineering



2) INSTALLING SOFTWARE



2) INSTALLING SOFTWARE

Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



↓ Windows

Windows 10, 11

User Installer [x64](#) [Arm64](#)
System
Installer [x64](#) [Arm64](#)
.zip [x64](#) [Arm64](#)
CLI [x64](#) [Arm64](#)



↓ .deb

Debian, Ubuntu

↓ .rpm

Red Hat, Fedora, SUSE

.deb [x64](#) [Arm32](#) [Arm64](#)
.rpm [x64](#) [Arm32](#) [Arm64](#)
.tar.gz [x64](#) [Arm32](#) [Arm64](#)
Snap [Snap Store](#)
CLI [x64](#) [Arm32](#) [Arm64](#)



↓ Mac

macOS 12.0+

.dmg [Intel chip](#) [Apple silicon](#) [Universal](#)
CLI [Intel chip](#) [Apple silicon](#)

2) INSTALLING SOFTWARE

The screenshot displays the Visual Studio Code interface with the Extensions Marketplace open. The search filter is set to 'MSSQL'. The 'SQL Server (mssql)' extension by Microsoft is highlighted. The extension's details page shows it has 9,045,655 downloads and a 4.5-star rating from 184 reviews. It is described as a tool to 'Design and optimize schemas for SQL Server, Azure SQL, and SQL Database in Fabric using a modern, lightweight extension built for developers'. The interface includes tabs for Details, Features, Changelog, Dependencies, and Extension Pack. A sidebar on the right provides installation details and marketplace information.

SQL Server (mssql)
Microsoft | 9,045,655 | 4.5 stars (184)
Design and optimize schemas for SQL Server, Azure SQL, and SQL Database in Fabric using a modern, lightweight extension built for developers

Disable | Uninstall | Auto Update

DETAILS | FEATURES | CHANGELOG | DEPENDENCIES | EXTENSION PACK

Roadmap 2026 | Report Bug | Request Feature | discussions 49 total | Build and Test (Unit + E2E) passing | codecov 74%

MSSQL extension for Visual Studio Code

The **MSSQL Extension for Visual Studio Code** is designed to empower developers by providing a seamless and modern database development experience. Our goal is to make SQL development more productive and intuitive by integrating essential features such as schema management, query execution, and AI-powered assistance.

Explore and Learn

Installation

Identifier	ms-mssql.mssql
Version	1.41.0
Last Updated	2 weeks ago
Size	17.78MB

Marketplace

Published	9 years ago
Last Released	2 weeks ago

2) INSTALLING SOFTWARE

The screenshot shows a web browser displaying a Microsoft Learn article. The address bar shows the URL: [learn.microsoft.com/en-us/sql/tools/visual-studio-code-extensions/mssql/mssql-extension-visual-studio-code?...](https://learn.microsoft.com/en-us/sql/tools/visual-studio-code-extensions/mssql/mssql-extension-visual-studio-code?view=sql-server-ver17). The page has a top navigation bar with 'Learn' and various dropdown menus. Below this is a 'SQL' section with more dropdowns and two buttons: 'Azure Portal' and 'Download SQL Server'. The left sidebar shows a tree view of the article's structure, with 'Overview' selected. The main content area has the title 'What is the MSSQL extension for Visual Studio Code?' and a sub-header 'The MSSQL extension for Visual Studio Code'. The text describes the extension's capabilities for connecting to and managing SQL databases within Visual Studio Code.

Version 

SQL Server 2025

Find by title

- Extended Features
- Visual Studio Code extensions
 - MSSQL extension
 - Overview**
 - Connect and query
 - Features
 - SQL bindings for Azure Functions
 - SQL Database Projects

Learn / SQL / SQL Server /

Ask Learn Focus mode

What is the MSSQL extension for Visual Studio Code?

Summarize this article for me

The [MSSQL extension for Visual Studio Code](#) supports developers building applications that use Azure SQL (including Azure SQL Database, Azure SQL Managed Instance, and SQL Server on Azure Virtual Machines), SQL database in Fabric, and SQL Server. It provides tools for connecting to databases, managing and designing schemas, exploring database objects, executing Transact-SQL (T-SQL) queries, and viewing query execution plans within Visual Studio Code.

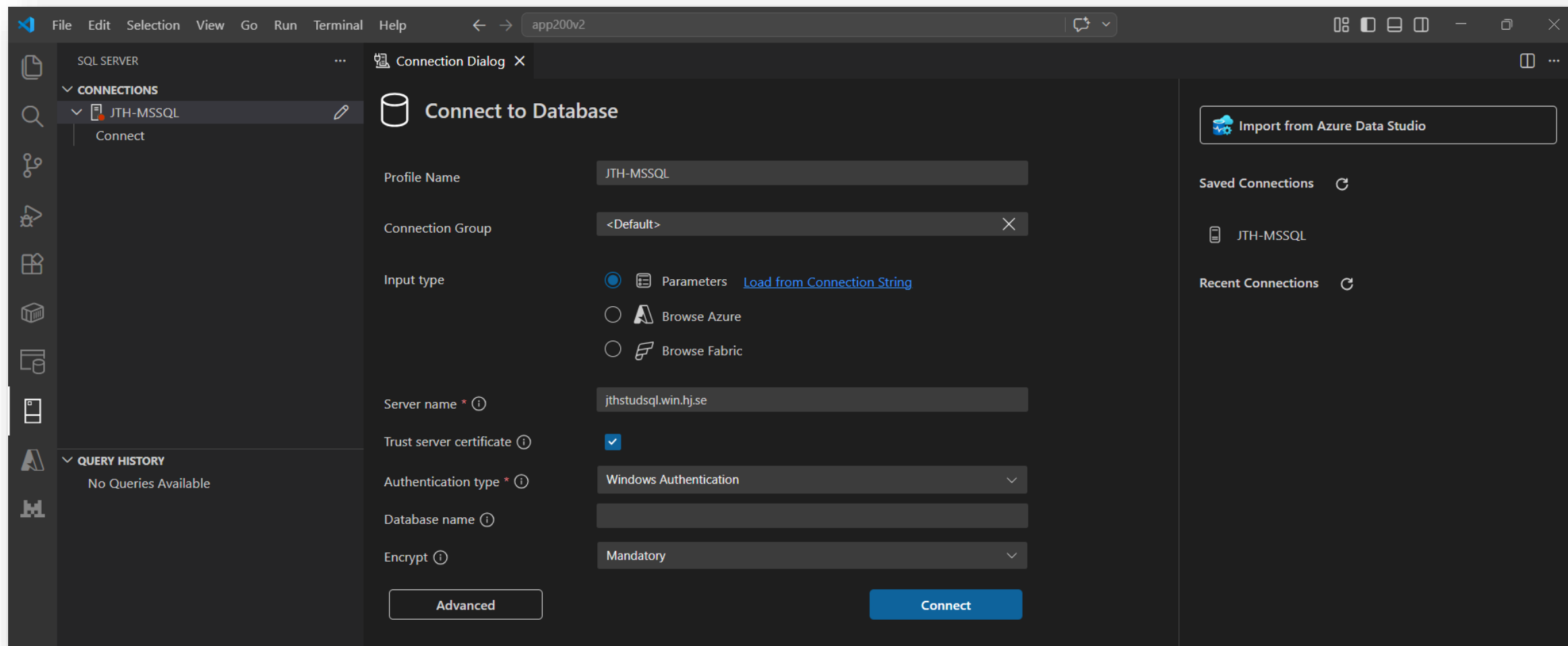
In this article

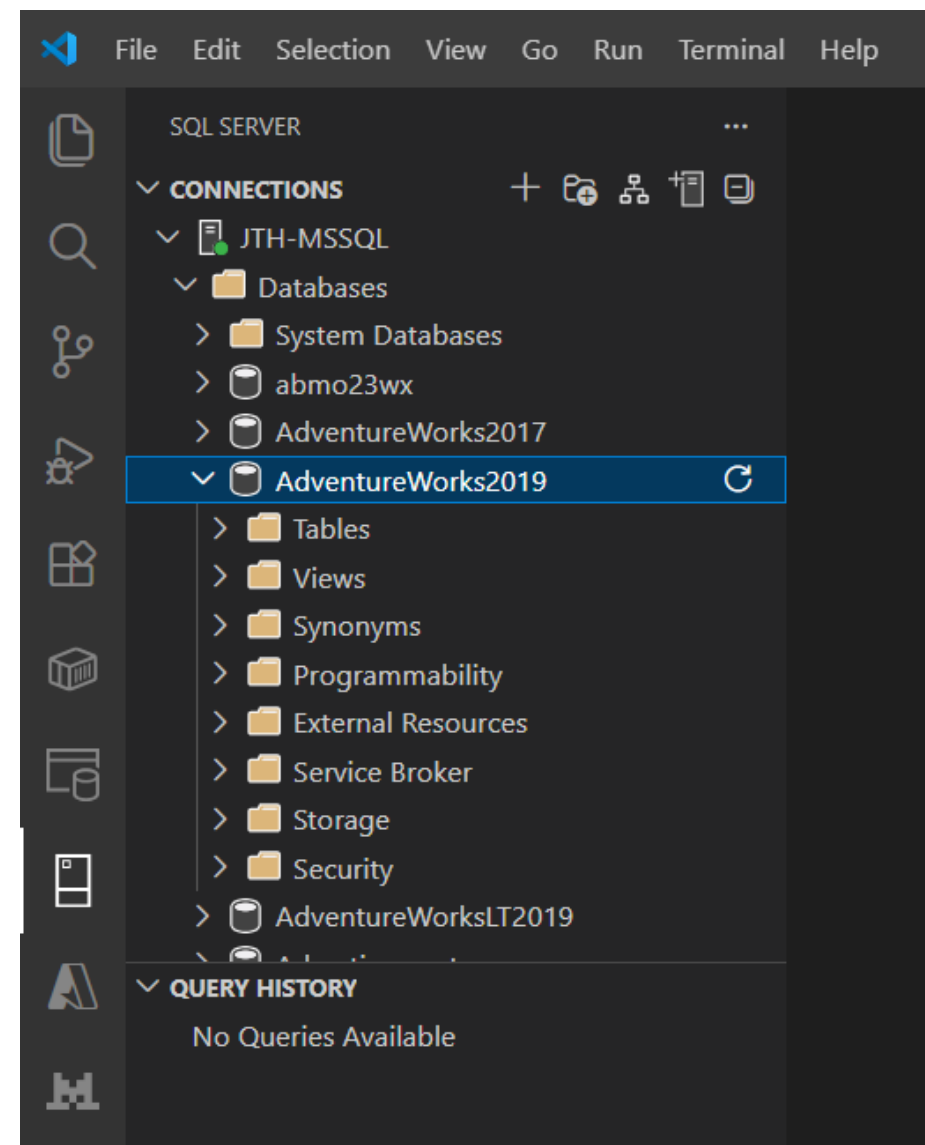
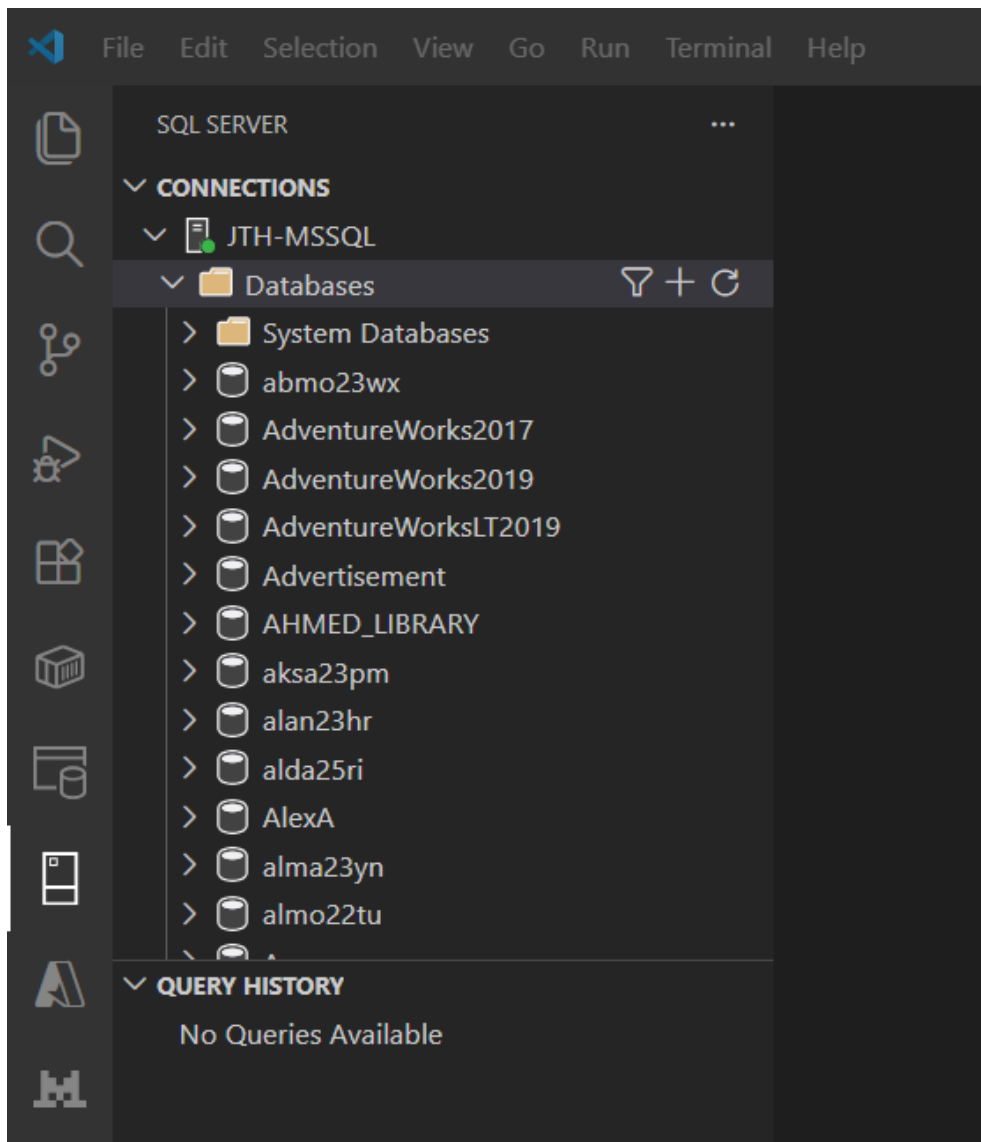
- Install the MSSQL Extension in Visual Studio Code
- Features
- Supported operating systems
- Offline installation
- Feedback and support
- Related content

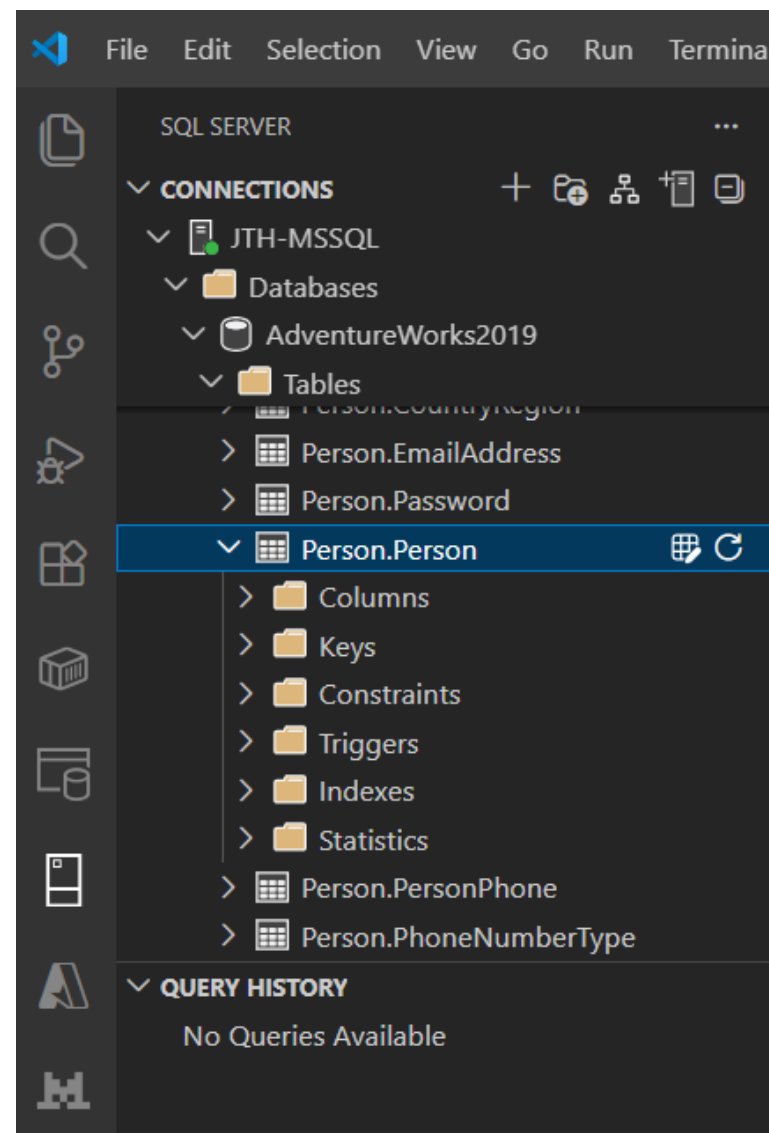
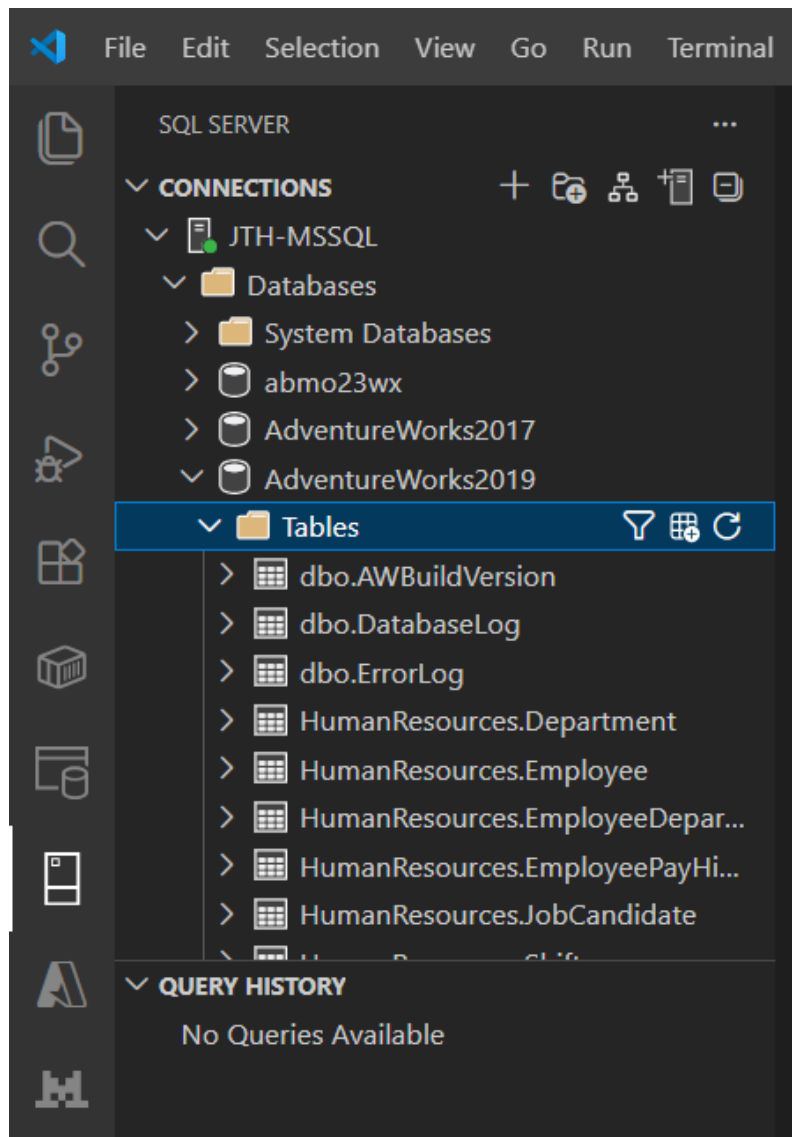
Was this page helpful?

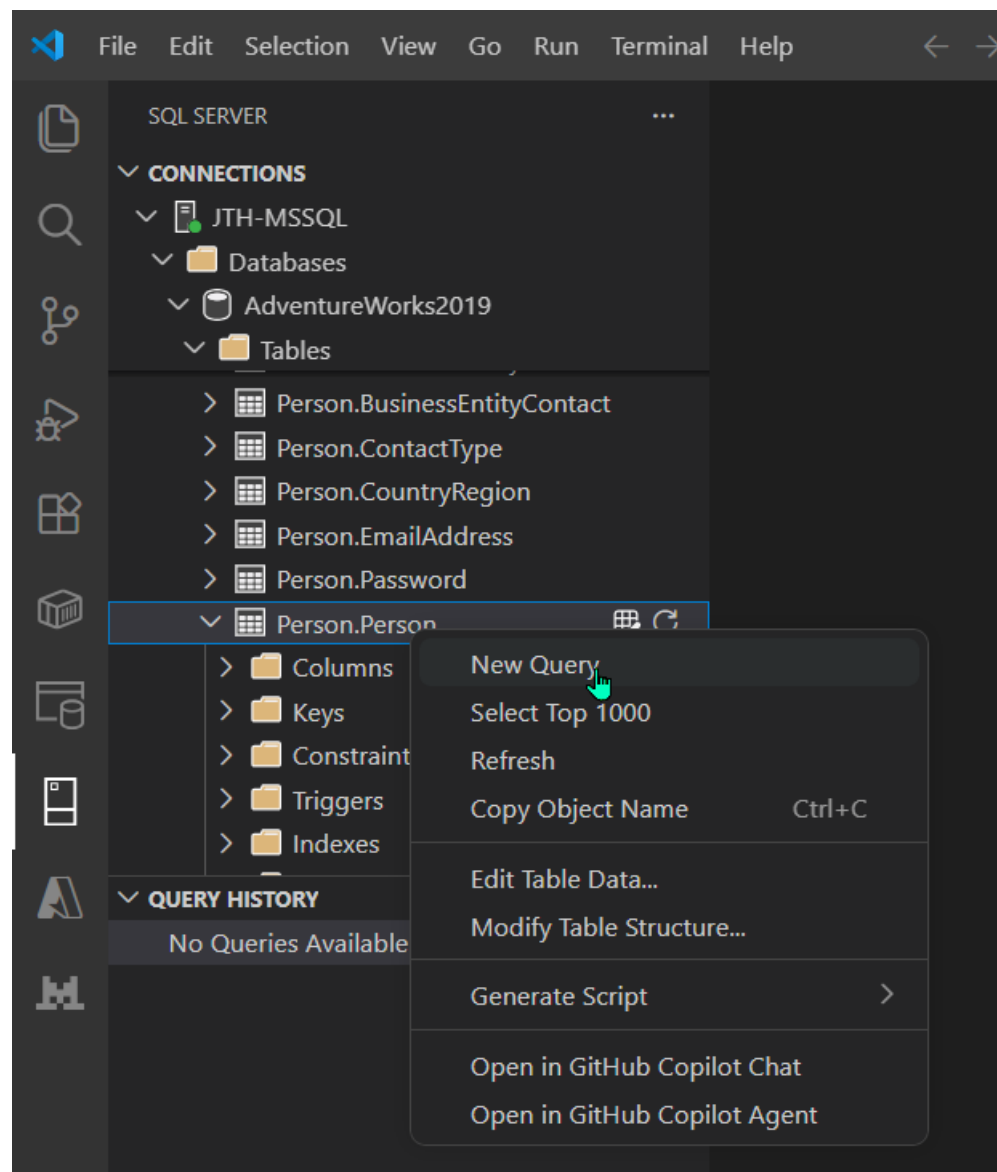
Yes No

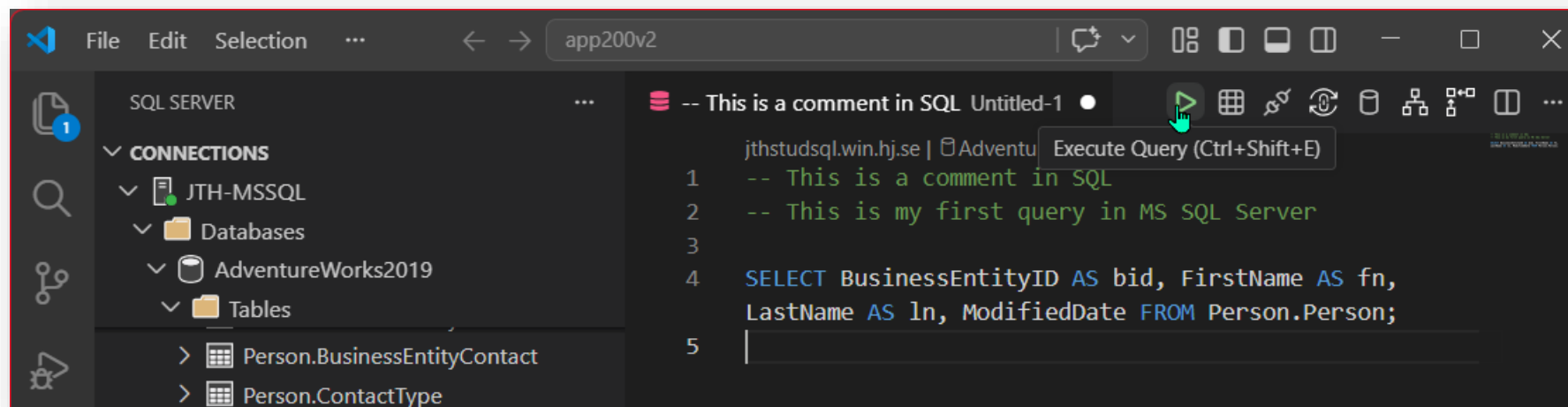
2) INSTALLING SOFTWARE











PROBLEMS OUTPUT DEBUG CONSOLE QUERY RESULTS ...

Results (1) Messages Open in New Tab

	bid	↑↓	🔍	fn	↑↓	🔍	ln	↑↓	🔍	ModifiedDate	↑↓	🔍
1	1			Ken			Sánchez			2009-01-07 00:00:00.000		
2	2			Terri			Duffy			2008-01-24 00:00:00.000		
3	3			Roberto			Tamburello			2007-11-04 00:00:00.000		
4	4			Rob			Walters			2007-11-28 00:00:00.000		
5	5			Gail			Erickson			2007-12-30 00:00:00.000		
6	6			Jossef			Goldberg			2013-12-16 00:00:00.000		
7	7			Dylan			Miller			2009-02-01 00:00:00.000		
8	8			Diane			Margheim			2008-12-22 00:00:00.000		
9	9			Gigi			Matthew			2009-01-09 00:00:00.000		

app200v2

SQL SERVER

CONNECTIONS

- JTH-MSSQL
 - Databases
 - AdventureWorks2019
 - Tables
 - Person.BusinessEntityContact
 - Person.ContactType
 - Person.CountryRegion
 - Person.EmailAddress
 - Person.Password
 - Person.Person
 - Columns
 - Keys
 - Constraints
 - Triggers
 - Indexes

QUERY HISTORY

No Queries Available

```
-- This is a comment in SQL
-- This is my first query in MS SQL Server

SELECT BusinessEntityID AS bid, FirstName AS fn,
       LastName AS ln, ModifiedDate FROM Person.Person;
```

PROBLEMS OUTPUT DEBUG CONSOLE QUERY RESULTS

Results (1) Messages Open in New Tab

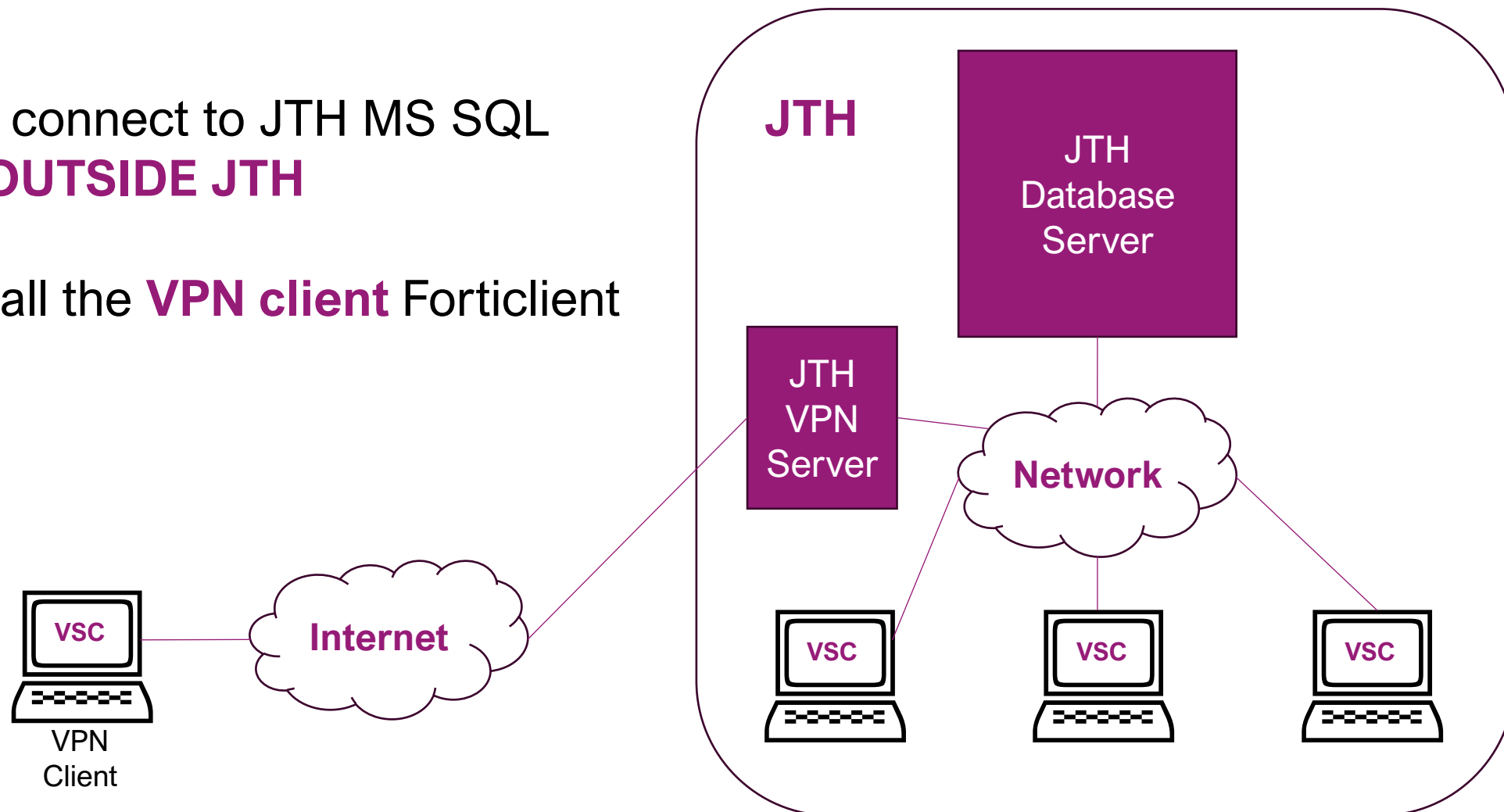
	bid	fn	ln	ModifiedDate
1	1	Ken	Sánchez	2009-01-07 00:00:00.000
2	2	Terri	Duffy	2008-01-24 00:00:00.000
3	3	Roberto	Tamburello	2007-11-04 00:00:00.000
4	4	Rob	Walters	2007-11-28 00:00:00.000
5	5	Gail	Erickson	2007-12-30 00:00:00.000
6	6	Jossef	Goldberg	2013-12-16 00:00:00.000
7	7	Dylan	Miller	2009-02-01 00:00:00.000
8	8	Diane	Margheim	2008-12-22 00:00:00.000
9	9	Gigi	Matthew	2009-01-09 00:00:00.000

SQLCMD: Off MSSQL 19972 rows affected 00:00:01.061 jthstudsql.win.hj.se AdventureWorks2019 Prettier

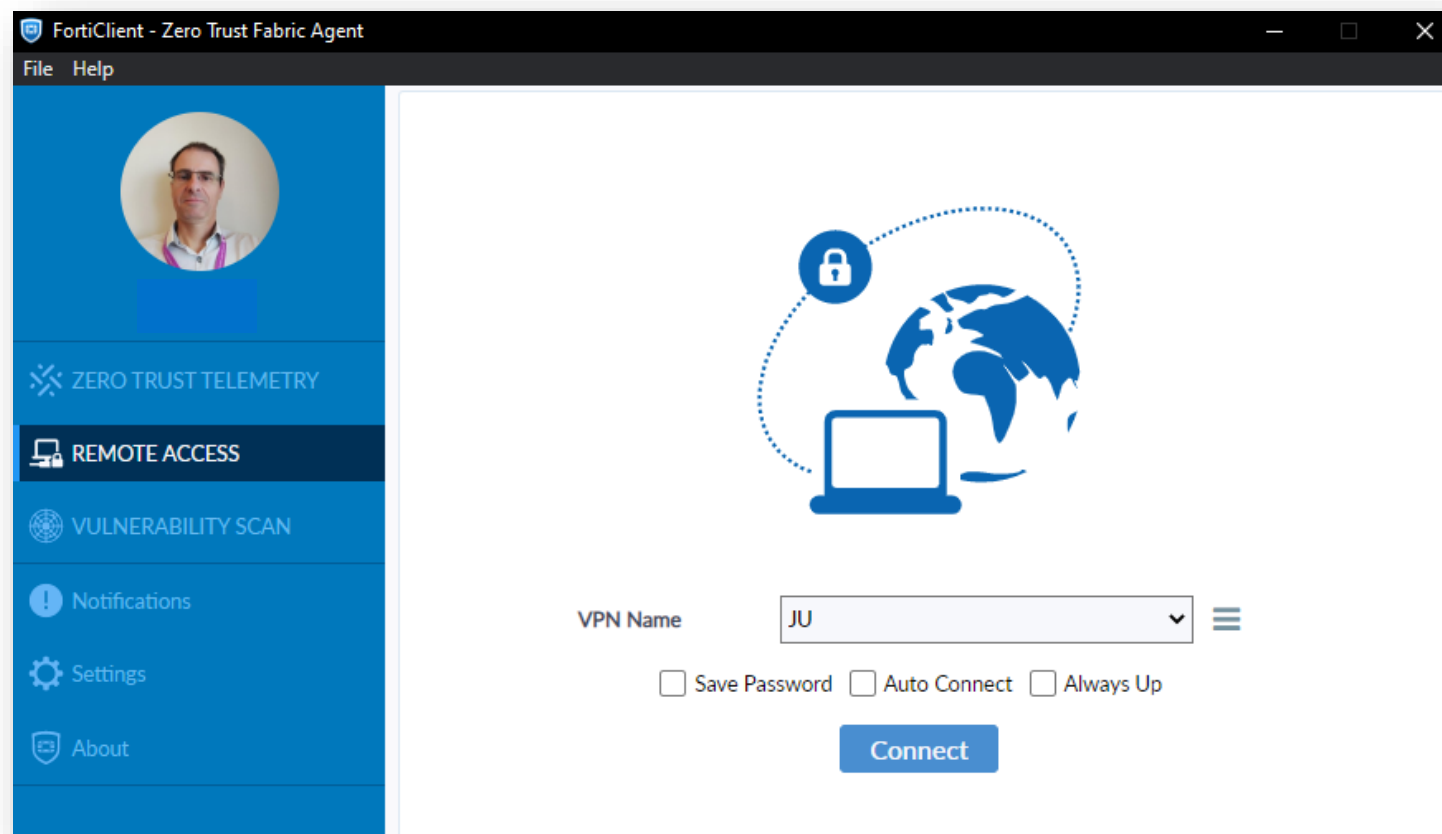
2) INSTALLING SOFTWARE

If you want to connect to JTH MS SQL Server from **OUTSIDE JTH**

You must install the **VPN client** Forticlient



2) INSTALLING SOFTWARE



2) INSTALLING SOFTWARE

View VPN Connection

VPN

SSL-VPNXML

Connection Name

JU

Description

Remote Gateway

vpn.ju.se

☒ Customize port

443

Single Sign On Settings

☒ Enable Single Sign On (SSO) for VPN Tunnel

☐ Use external browser as user-agent for saml user authentication

☐ Enable auto-login with Azure Active Directory

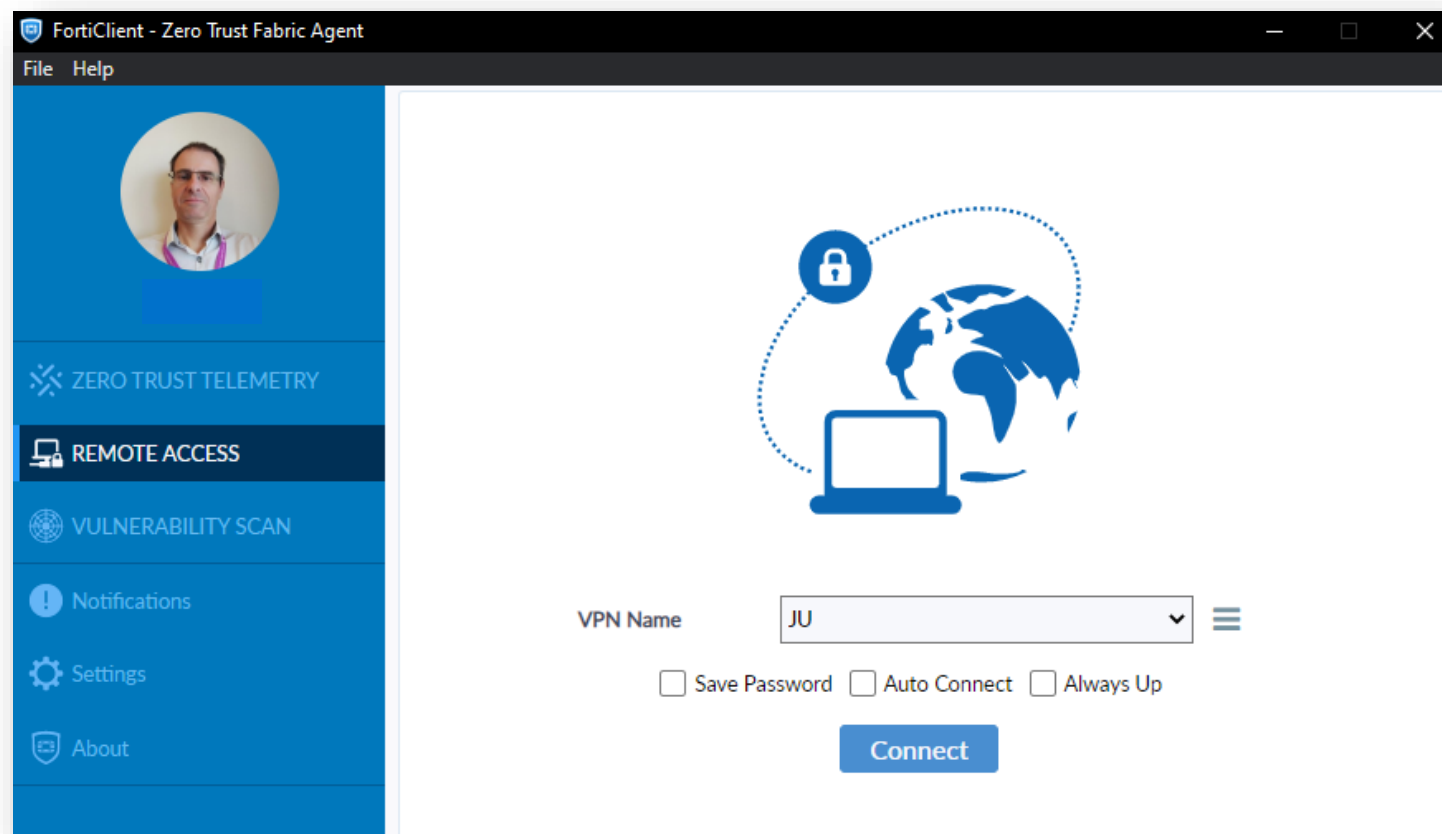
Client Certificate

None

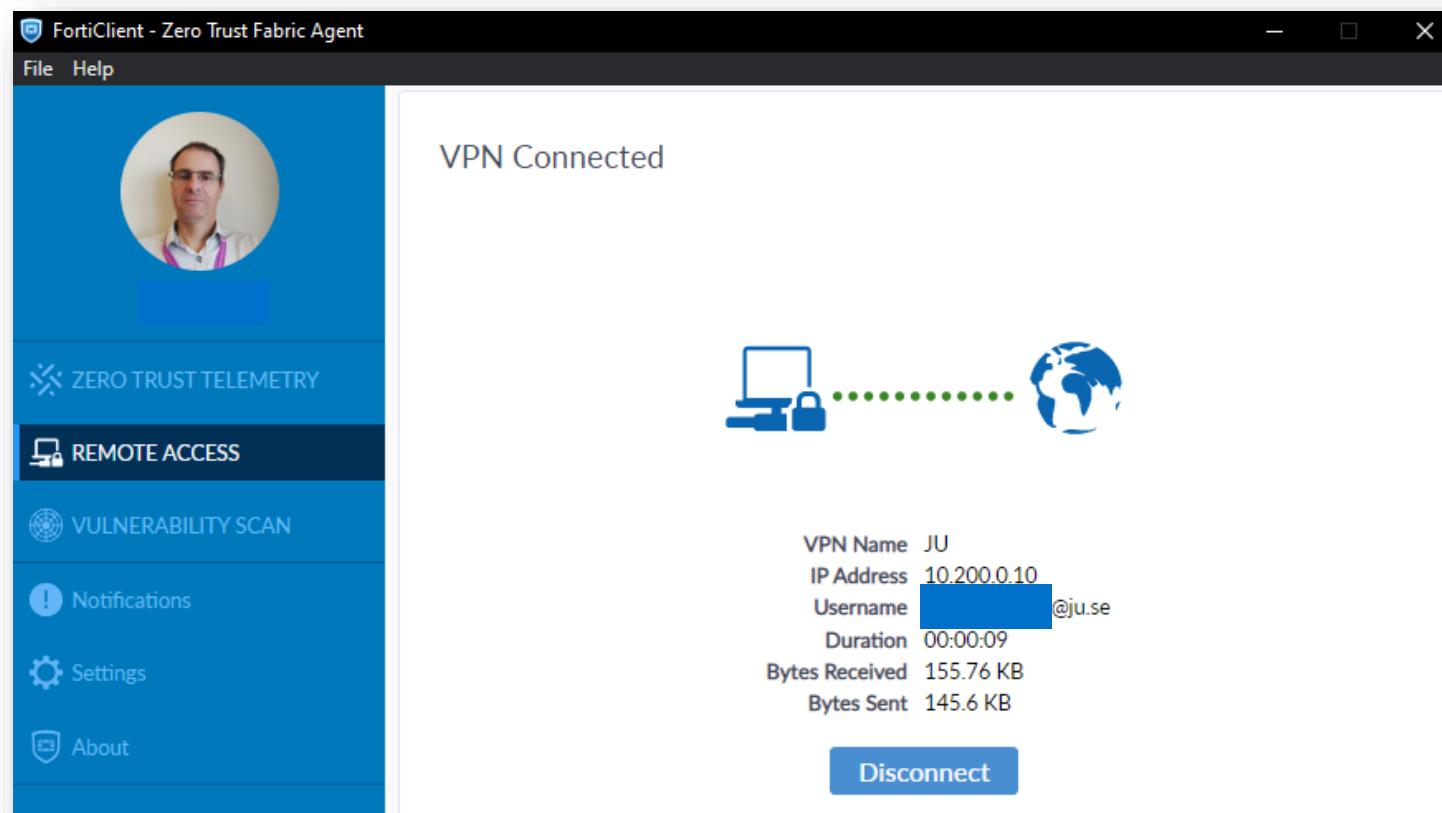
☐ Enable Dual-stack IPv4/IPv6 address

Cancel

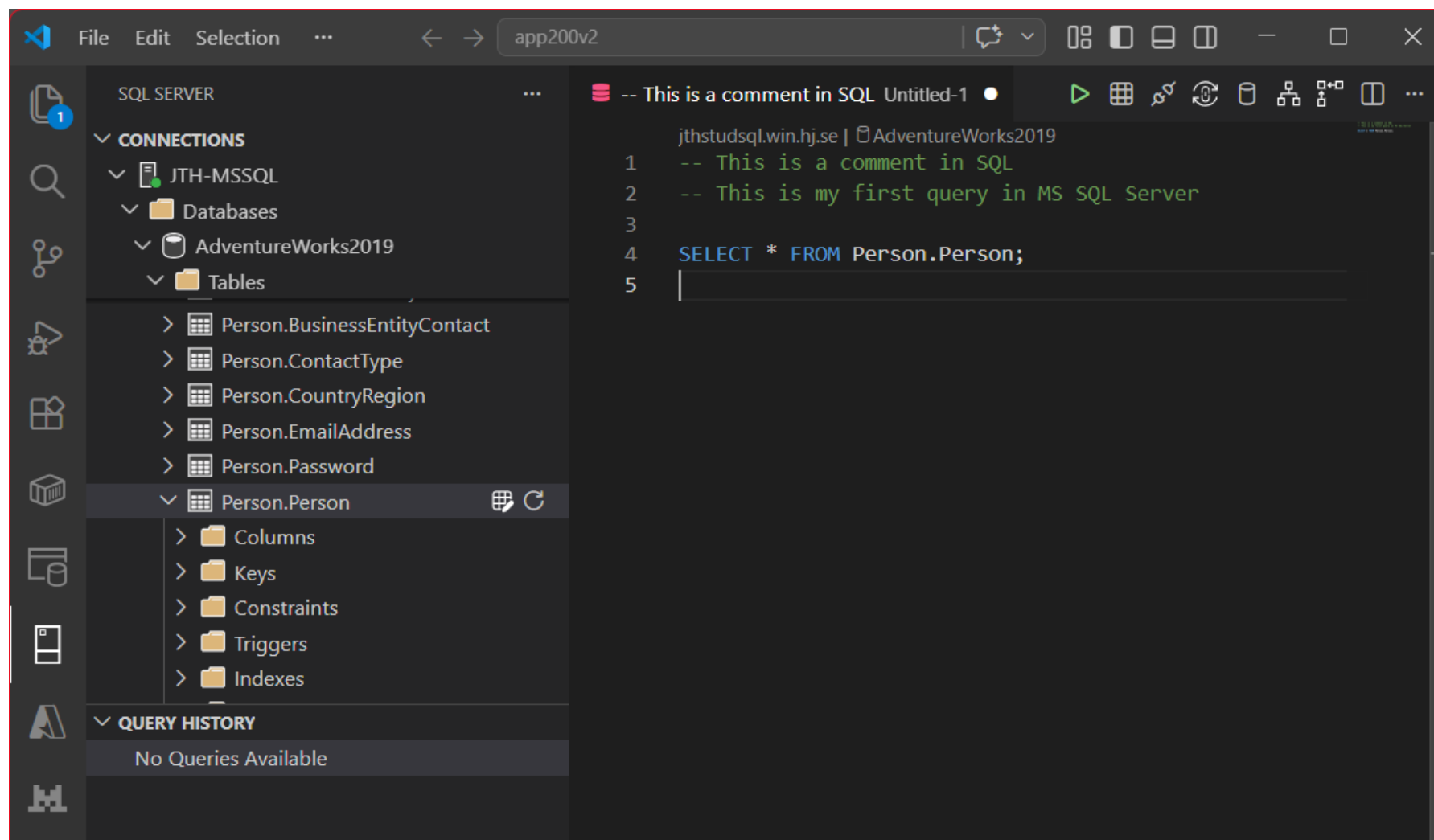
2) INSTALLING SOFTWARE



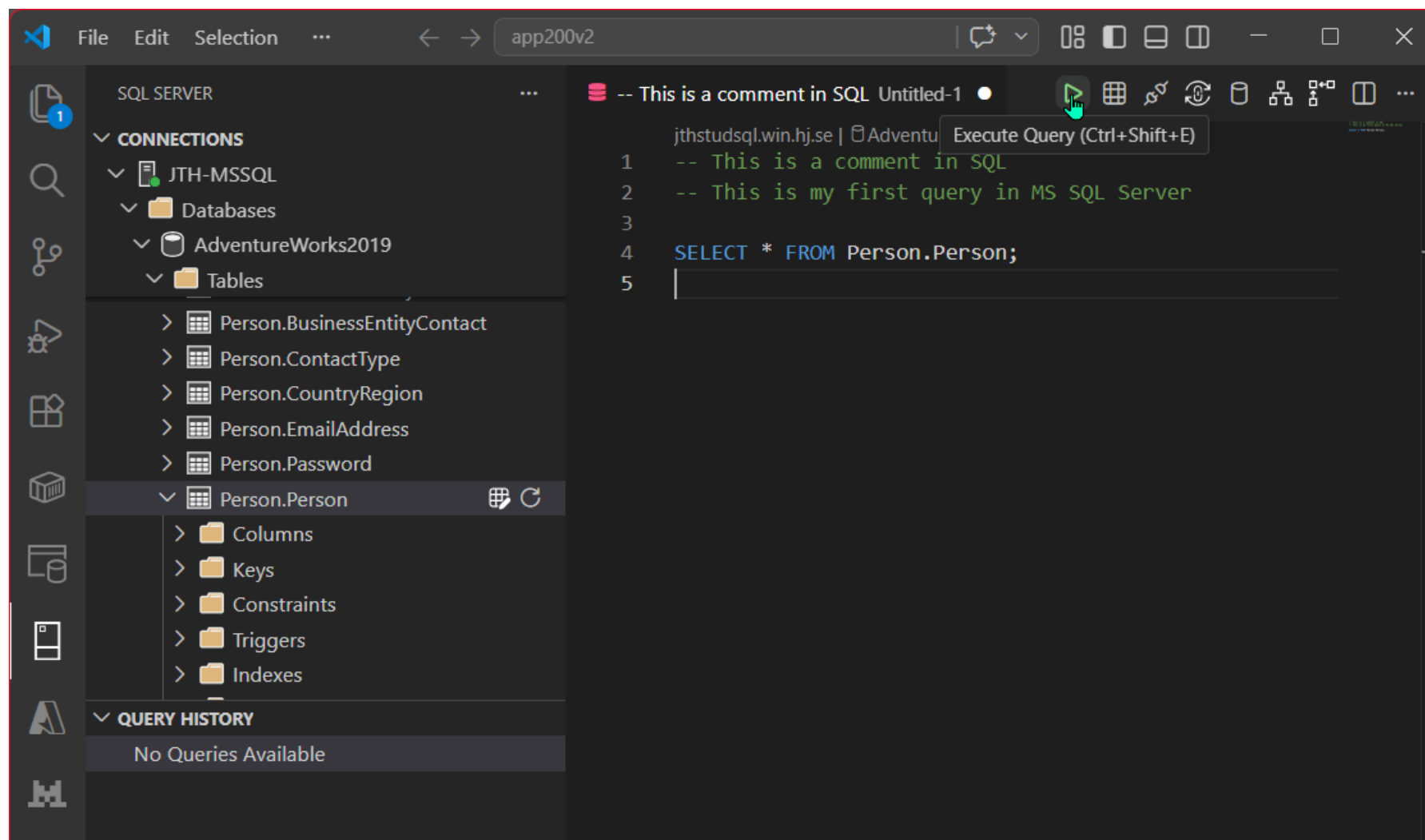
2) INSTALLING SOFTWARE



2) INSTALLING SOFTWARE



2) INSTALLING SOFTWARE



SQL SERVER

CONNECTIONS

JTH-MSSQL

Databases

AdventureWorks2019

Tables

Person.BusinessEntityContact

Person.ContactType

Person.CountryRegion

Person.EmailAddress

Person.Password

Person.Person

Columns

Keys

Constraints

Triggers

Indexes

QUERY HISTORY

No Queries Available

-- This is a comment in SQL

Untitled-1

jthstudsql.win.hj.se | AdventureWorks2019

-- This is a comment in SQL

-- This is my first query in MS SQL Server

SELECT * FROM Person.Person;

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

QUERY RESULTS

AZURE

Results (1)

Messages

Open in New Tab

	BusinessEntityID	PersonType	NameStyle	Title	FirstName	MiddleName	LastName	Suffix	EmailP
1	1	EM	0	NULL	Ken	J	Sánchez	NULL	0
2	2	EM	0	NULL	Terri	Lee	Duffy	NULL	1
3	3	EM	0	NULL	Roberto	NULL	Tamburello	NULL	0
4	4	EM	0	NULL	Rob	NULL	Walters	NULL	0
5	5	EM	0	Ms.	Gail	A	Erickson	NULL	0
6	6	EM	0	Mr.	Jossef	H	Goldberg	NULL	0
7	7	EM	0	NULL	Dylan	A	Miller	NULL	2
8	8	EM	0	NULL	Diane	L	Margheim	NULL	0
9	9	EM	0	NULL	Gigi	N	Matthew	NULL	0

<

0

0

Ln 5, Col 1

Spaces: 4

UTF-8

CRLF

{ }

SQL

SQLCMD: Off

MSSQL

19972 rows affected

00:00:10.759

jthstudsql.win.hj.se

AdventureWorks2019

Prettier

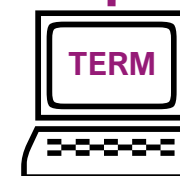
2) INSTALLING SOFTWARE

In case it does not work, let's have a
PLAN B

Let's use **MariaDB** running in a **Docker container** (see end of lecture 00)

JTH

Docker container
MariaDB



TIPS AND TRICKS

In life, of course, **be positive, B+**

But please, always have a **PLAN B**

3) DATA DEFINITION LANGUAGE (DDL)

In Microsoft SQL Server, a **database is a structured collection of data organized into tables**, views, and other objects. It serves as a container for storing, managing, and retrieving data efficiently[1][3].

- Key Characteristics:

1. **Relational Model:** SQL Server databases are built on the relational model, where data is organized into tables with rows and columns[2][4].
2. **ACID Compliance:** Databases in SQL Server adhere to ACID (Atomicity, Consistency, Isolation, Durability) principles, ensuring data integrity and reliable transaction processing[3].
3. **Scalability:** SQL Server can handle databases ranging from small single-machine applications to large Internet-facing applications with many concurrent users[7].
4. **Object Organization:** Databases contain various objects such as tables, views, stored procedures, indexes, and triggers, which are logically grouped into schemas[6].
5. **Physical Storage:** Physically, SQL Server databases are stored as a set of 8 KB pages, typically in the PRIMARY default filegroup[4].

- Components and Features:

- **Tables:** The fundamental storage units for data, organized in rows and columns[2][4].
- **Schemas:** Logical containers for database objects, providing organization and security boundaries[6].
- **Query Language:** Uses Transact-SQL (T-SQL), Microsoft's proprietary extension of SQL, for data manipulation and management[1][5].
- **Security:** Implements various security measures, including user authentication and schema-level permissions[6].
- **System Tables:** Special tables that store information about the SQL Server instance configuration and object information[2].

SQL Server databases form the core of Microsoft's relational database management system, providing a robust platform for data storage, retrieval, and management in various applications and enterprise environments.

Citations:

[1] <https://www.techtarget.com/searchdatamanagement/definition/SQL-Server>

[3] <https://www.actian.com/what-is-sql-server/>

[5] <https://hasura.io/learn/database/microsoft-sql-server/what-is-mssql/>

[7] https://en.wikipedia.org/wiki/Microsoft_SQL_Server

[9] <https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-sql-database>

[2] <https://www.linkedin.com/pulse/sql-server-table-structure-overview-eng-ahmad-yaseen>

[4] <https://www.sqlshack.com/sql-server-table-structure-overview/>

[6] <https://www.tutorialsteacher.com/sqlserver/database-schema>

[8] <https://learn.microsoft.com/en-us/sql/relational-databases/security/authentication-access/create-a-database-schema?view=sql-server-ver16>

[10] <https://study.com/academy/lesson/what-is-microsoft-sql-server.html>

3) DATA DEFINITION LANGUAGE (DDL)

How to create a database?

-- SQL comment, create a new DB

CREATE DATABASE myBank;

How to **delete** a database?

-- SQL comment, delete DB (and all content!)

DROP DATABASE myBank;

3) DATA DEFINITION LANGUAGE (DDL)

In Microsoft SQL Server (MSSQL), a **schema** is a **logical container for database objects** such as tables, views, stored procedures, functions, and indexes [3][6]. It serves as a namespace that organizes and groups related database objects, providing a way to manage security and separate different areas of concern within a single database [6].

- Key characteristics of schemas in MSSQL include:

1. **Ownership**: A schema can be owned by a database user, role, or application role [3].
2. **Default** schema: The 'dbo' schema is the default for newly created databases [3].
3. **Multiple** schemas: A single database can contain multiple schemas [6].
4. **Security** management: Schemas allow for granular control over user access permissions to database objects [1][6].
5. **Naming convention**: Objects within a schema are referenced using the format [schemaName].[objectName] [6].

- Schemas in MSSQL offer several benefits:

- Improved organization of database objects
- Enhanced security through schema-level permissions
- Simplified management of large databases with many objects
- Ability to transfer ownership of multiple objects simultaneously

It's important to note that while the term "schema" in MSSQL refers to this specific container concept, it differs from the broader definition of a database schema, which describes the overall structure and relationships of the entire database [5].

Citations:

[1] <https://www.cockroachlabs.com/blog/database-schema-beginners-guide/>

[3] <https://www.tutorialsteacher.com/sqlserver/database-schema>

[5] <https://stackoverflow.com/questions/45141673/which-of-definitions-of-database-schema-is-correct>

[7] <https://learn.microsoft.com/en-us/sql/relational-databases/security/authentication-access/create-a-database-schema?view=sql-server-ver16>

[8] <https://www.sqlshack.com/introducing-schema-documentation-in-sql-server/>

[2] <https://www.simplilearn.com/tutorials/sql-tutorial/schema-in-sql>

[4] <https://www.sqlshack.com/a-walkthrough-of-sql-schema/>

[6] https://www.reddit.com/r/learnprogramming/comments/15svdmo/can_someone_explain_what_a_database_schema_is/

3) DATA DEFINITION LANGUAGE (DDL)

How to create a schema?

-- SQL comment, create a new schema

CREATE SCHEMA Person;

How to **delete** a schema?

-- SQL comment, delete schema (and all content!)

DROP SCHEMA Person;

3) DATA DEFINITION LANGUAGE (DDL)

How to create a table?

-- SQL comment, create a new schema

```
CREATE TABLE Person (  
    pId INT PRIMARY KEY IDENTITY(1,1),  
    pFName NVARCHAR(60),  
    pLName NVARCHAR(50),  
    pBirthDate DATE,  
    pEmail NVARCHAR(100),  
    pPhone NVARCHAR(20)  
);
```

How to **delete** a table?

-- SQL comment, delete schema (and all content!)

```
DROP TABLE Person;
```

We NEED datatypes!!!

Data type	Length	Description
bigint	8	Integer from -2^{63} (-9 223 372 036 854 775 808) to $2^{63}-1$ (9 223 372 036 854 775 807)
int	4	Integer from -2^{31} (-2 147 483 648) to $2^{31}-1$ (2 147 483 647)
smallint	2	Integer from -2^{15} (-32 768) to $2^{15}-1$ (32 767)
tinyint	1	Integer from 0 to 255
bit	1 bit	Integer 0 or 1.
decimal(precision, scale)	5-17	Numeric data type with fixed precision and scale (precision 1-38, 18 by default and scale 0-p, 0 by default).
numeric	5-17	The same as decimal data type.
money	8	Financial data type from -2^{63} (-922 337 203 685 477.5808) to $2^{63}-1$ (922 337 203 685 477.5807) with the precision of one ten-thousandth unit.
smallmoney	4	Financial data type from -2^{31} (-214 748.3648) to $2^{31}-1$ (214 748.3647) with the precision of one ten-thousandth unit.
float(n)	4-8	Numeric data type with float precision, where n is the number of mantis bits (1-24, precision of 7 digits, size of 4 bytes and 25-53, precision of 15 digits and size of 8 bytes).
real	4	Numeric data type with float precision that is defined as a float(24) .
datetime	8	Data type representing date and time from 1.1.1753 to 31.12.9999 with the precision about 3 ms. Values are rounded to .000, .003 and .007.
smalldatetime	4	Data type representing date and time from 1.1.1900 to 6.6.2079 with the precision of 1 minute. Values up to 29.998 are rounded down and values from 29.999 are rounded up to the nearest minute.

char	n	Text string of fixed length (maximum length: 8000 characters)
varchar	n	Text string of variable length (maximum length: 8000 characters)
text	n	Text string of variable length (maximum length: $2^{31}-1 = 2\,147\,483\,647$ characters)
nchar	$2*n$	Unicode text string of fixed length (maximum length: 4000 characters)
nvarchar	$2*n$	Unicode text string of variable length (maximum length: 4000 characters)
ntext	$2*n$	Unicode text string of variable length (maximum length: $2^{30}-1 = 1\,073\,741\,823$ characters)
binary	n+4	Binary data of fixed length (maximum length: 8000 bytes)
varbinary	n+4	Binary data of variable length (maximum length: 8000 bytes)
image	n	Binary data of variable length (maximum length: $2^{31}-1 = 2\,147\,483\,647$ bytes)
cursor		For storing the reference to cursors in the variable or in a procedure (no in the <code>CREATE TABLE</code> method).
sql_variant		For storing value of another type (no <code>text</code> , <code>ntext</code> , <code>image</code> , <code>timestamp</code> , <code>sql_variant</code>) of max. length to 8016 bytes. ODBC doesn't fully support this data type.
table		For storing the query result for the later usage.
timestamp	8+4	Data type generates automatically binary numbers, unique in the database, used usually to the rows identification. There can be only column of this data type in the table.
uniqueidentifier		Data type for storing GUID (new by means of the <code>NEWID</code> function or existing from the string in the form <code>xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx</code> , for example <code>6F9619FF-8B86-D011-B42D-00C04FC964FF</code>).

3) DATA DEFINITION LANGUAGE (DDL)

How to modify/alter a table?

-- SQL comment, alter a column type

ALTER TABLE Person

ALTER COLUMN phone NVARCHAR(30);

-- MSSQL Server: add a column

ALTER TABLE Person

ADD birthdate DATE;

-- MSSQL Server: remove a column

ALTER TABLE Person

DROP COLUMN middlename;

3) DATA DEFINITION LANGUAGE (DDL)

How to modify/alter a table?

-- MSSQL Server: rename a column

EXEC sp_rename 'phone', 'newphone', 'COLUMN';

-- MSSQL Server: rename a table

EXEC sp_rename 'Oldemployees', 'Newemployees';

3) DATA DEFINITION LANGUAGE (DDL)

Constraints:

-- MSSQL Server: constraint on a primary key

ALTER TABLE Employees

ADD CONSTRAINT PK_employeeid PRIMARY KEY (employeeid);

-- MSSQL Server: constraint on a foreign key

ALTER TABLE Orders

**ADD CONSTRAINT FK_customerid FOREIGN KEY (customerid) REFERENCES
Customers(customerid);**

-- MSSQL Server: remove a constraint

ALTER TABLE Employees

DROP CONSTRAINT PK_employeeid;

3) DATA DEFINITION LANGUAGE (DDL)

Using constraints:

-- MSSQL Server: constraint on delete

ALTER TABLE Orders

ADD CONSTRAINT FK_customerid

FOREIGN KEY (customerid) REFERENCES Customers(customerid)

ON DELETE NO ACTION;

NO ACTION (default)

If you try to delete a customer who has orders, the deletion will fail.

CASCADE

If you delete a customer, all their orders are also deleted.

DANGER!

Can be NULL!

SET NULL

If you delete a customer, the customerid in their orders is set to NULL.

Has a DEFAULT value!

SET DEFAULT

If you delete a customer, the customerid in their orders is set to a predefined default value.

3) DATA DEFINITION LANGUAGE (DDL)

Using constraints:

-- MSSQL Server: define a default value during the creation of the table

```
CREATE TABLE Orders (  
    OrderID INT PRIMARY KEY,  
    CustomerID INT,  
    OrderDate DATE,  
    CONSTRAINT DF_CustomerID DEFAULT 0 FOR CustomerID  
);
```

-- MSSQL Server: define a default value using a constraint

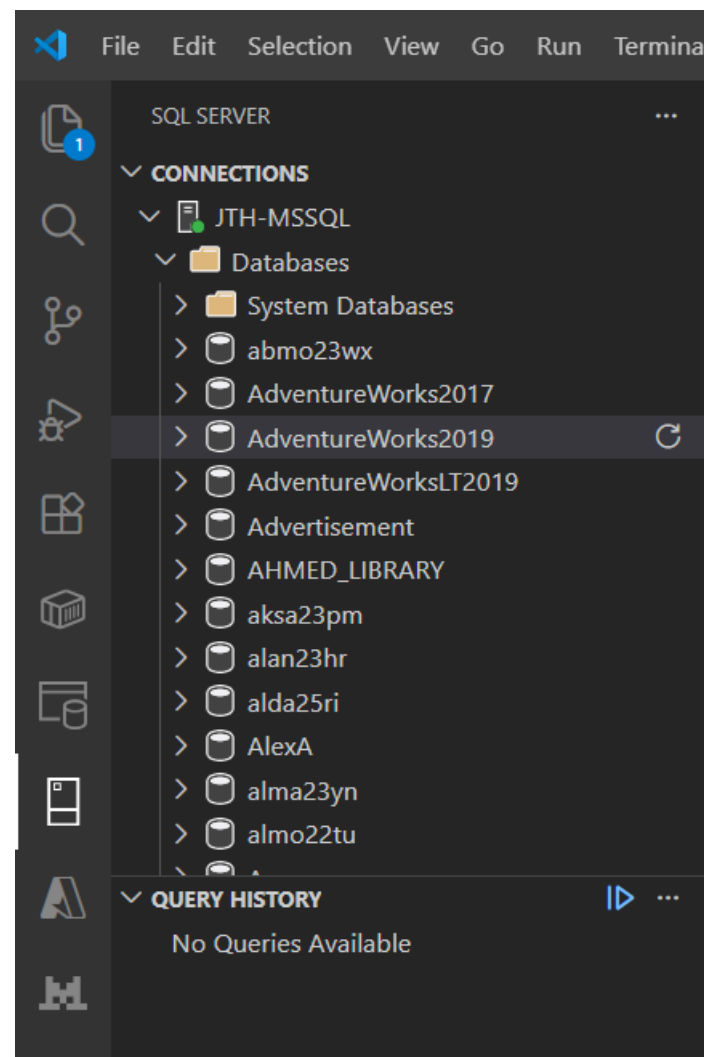
```
ALTER TABLE Orders  
ADD CONSTRAINT DF_CustomerID DEFAULT 0 FOR CustomerID;
```

3) DATA DEFINITION LANGUAGE (DDL)

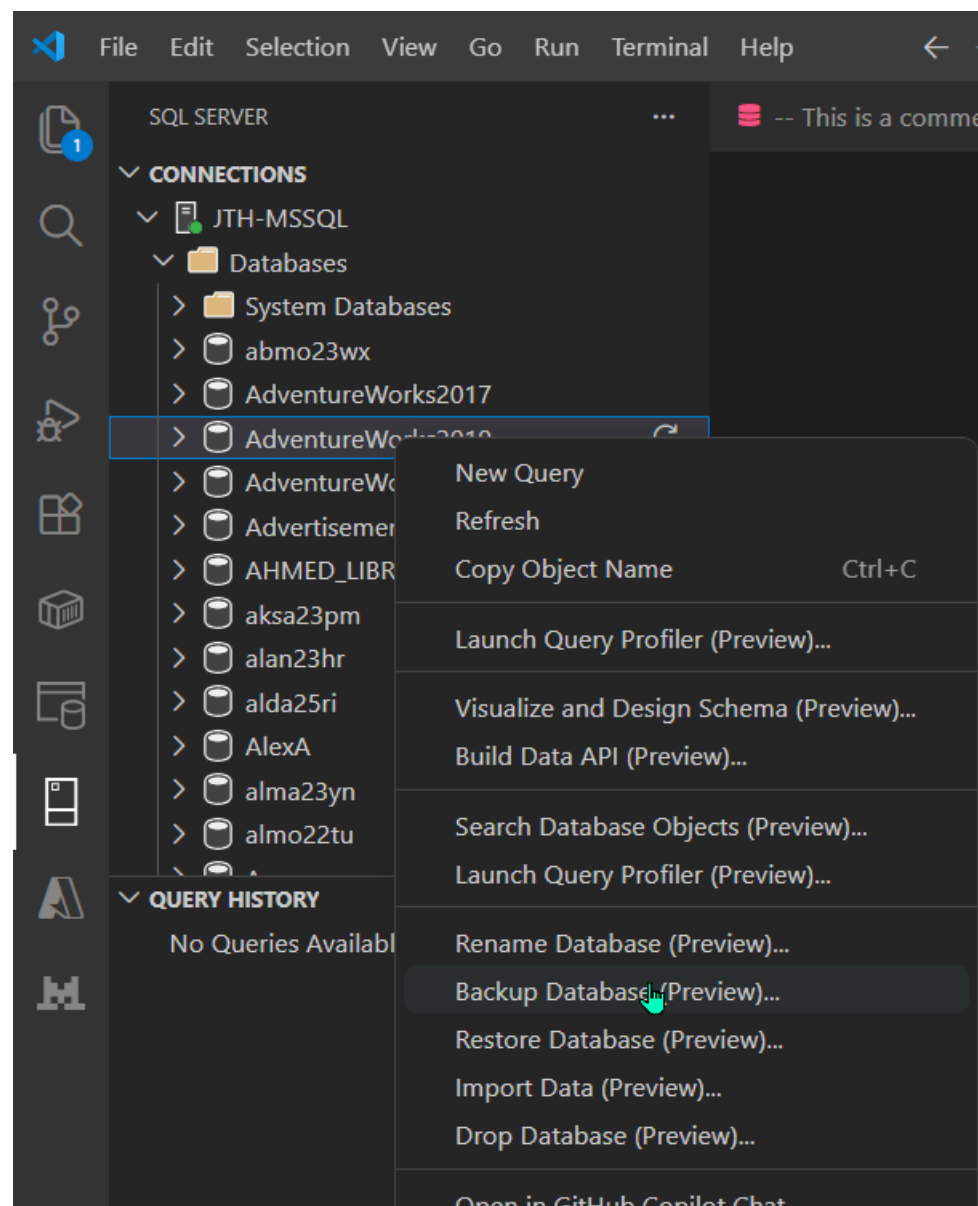
Before changing **anything** on the database structure,
do a **BACKUP** to ensure you saved the data!

3) DATA DEFINITION LANGUAGE (DDL)

Database BACKUP:



Database BACKUP:



Database BACKUP:

 **Backup Database (Preview) - AdventureWorks2019**

Backup Name
AdventureWorks2019-Full-2026-04-05T06-24-50.bak

Backup Type
Full

Copy-only Backup ☐

Backup Location
☒  Save to Disk
☐  Save to URL

Backup Files *

 **New File** 

Folder Path D:\MSSQL\Backup

File Name AdventureWorks2019-Full-2026-(

Create New

Choose Existing

Advanced

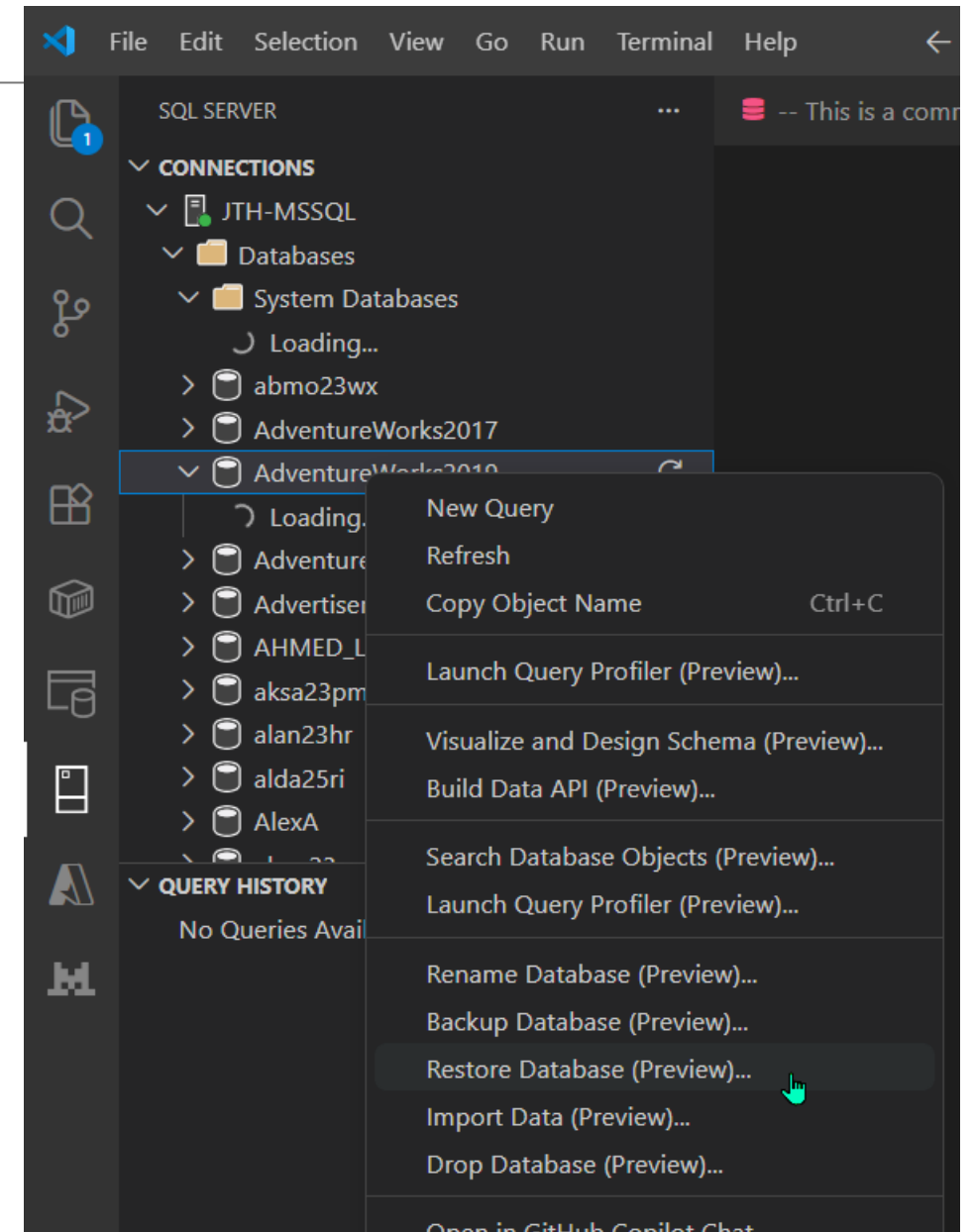
Help

Script

Cancel

Backup

Database **RESTORE**:





Restore Database (Preview) - jthstuds.sql.win.hj.se

Backup Location

☐ Database

☒ Backup File

☐ URL

Backup Files *

Browse files

! Please choose at least one backup file

Target Database

MyBank



Please choose a backup file to load restore plan

Advanced

Help

Script

Cancel

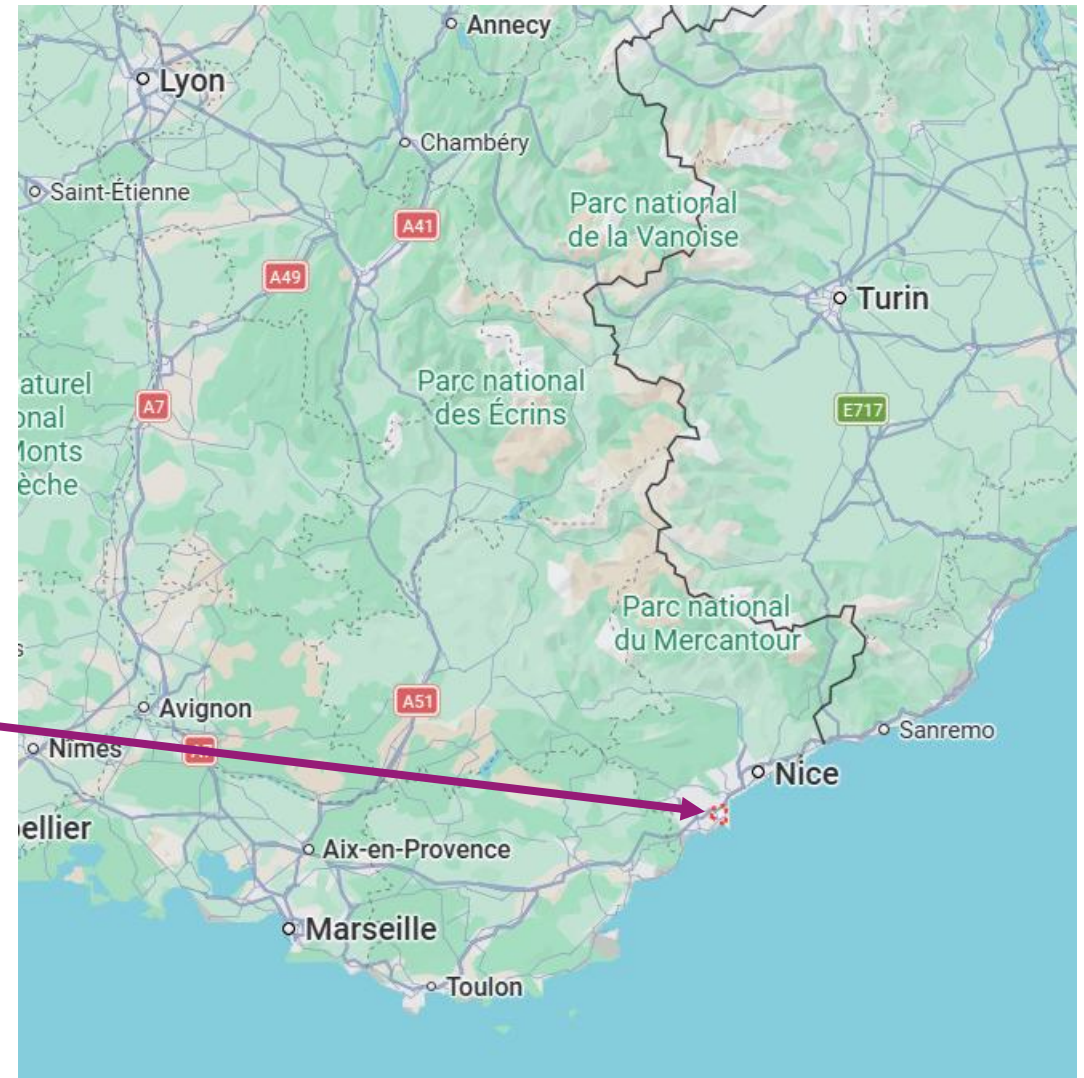
Restore

TIPS AND TRICKS

Are you **positive** today?

To be positive, we need sun,
sea, waves, ...

Let's go to **Antibes!**



Le cap d'Antibes, 2026-01





Be positive!

B+



4) DATA MANIPULATION LANGUAGE (DML)

DML stands for **Data Manipulation Language**, a subset of SQL (Structured Query Language) used to manipulate data stored in databases [1][2]. It encompasses commands for performing **CRUD (Create, Read, Update, Delete)** operations on database tables [1][3].

- The main DML commands include:

1. **INSERT**: Adds new data to the database [2][4]
2. **SELECT**: Retrieves data from the database (often considered part of DML, though technically part of DQL)[3][4]
3. **UPDATE**: Modifies existing data in the database [2][4]
4. **DELETE**: Removes data from the database [2][4]

DML is essential for effective data management, **allowing users to interact with and modify data efficiently** across various database management systems (DBMS) such as Microsoft SQL Server, MySQL, PostgreSQL, and Oracle [1].

- Key Characteristics of DML:

- It focuses on **manipulating the data** within database structures, **not the structures** themselves [4].
- DML statements are **transactional**, meaning they can be rolled back if necessary [4].
- These commands are typically executed by application developers or end-users during normal database operations [4].

By using DML commands, database administrators and users can effectively manage and update the information stored in their databases, ensuring data accuracy and relevance.

Citations:

[1] <https://www.datasunrise.com/knowledge-center/dml-data-manipulation-language/>

[3] https://en.wikipedia.org/wiki/Data_manipulation_language

[5] <https://stackoverflow.com/questions/2578194/what-are-ddl-and-dml>

[7] <https://web.csulb.edu/colleges/coe/cecs/dbdesign/dbdesign.php?page=sql%2Fddl-dml.php>

[2] <https://satoricyber.com/glossary/dml-data-manipulation-language/>

[4] <https://www.almabetter.com/bytes/tutorials/sql/dml-ddl-commands-in-sql>

[6] <https://www.datacamp.com/tutorial/sql-dml-commands-mastering-data-manipulation-in-sql>

Answer from Perplexity: pplx.ai/share

4) DATA MANIPULATION LANGUAGE (DML)

CRUD

Create

INSERT

Read

SELECT

Update

UPDATE

Delete

DELETE

4) DATA MANIPULATION LANGUAGE (DML)

How to insert a new row?

-- Insert a new row

INSERT INTO Person

(pFName, pLName, pEmail, pPhone)

VALUES

("Jérôme", "Landré", "j.l@ju.se", "070605040302")

How to **delete** a row?

-- Delete a row

DELETE FROM Person

WHERE pid=123;

4) DATA MANIPULATION LANGUAGE (DML)

How to **update/modify** a row?

-- Update/Modify a row

```
UPDATE Person
SET pFname="Jérôme", pPhone="0607080910"
WHERE pid=123;
```

-- Update/Modify many rows

```
UPDATE Person
SET salary=salary*1.01 -- increase by 1%
WHERE pBirthDate>"2000-01-01"; -- young persons
```

How to read a row/rows?

-- Read one row

```
SELECT * FROM Person
WHERE pid=123;
```

-- Read many rows

```
SELECT * FROM Person
WHERE pBirthDate>"2000-01-01";
```

5) DEMONSTRATION

Demonstration on MSSQL Server

5) DEMONSTRATION

Remember that **SQL** queries are just **TEXT** commands.

ANY QUESTIONS?